

Demonstrating SkinHaptics: Exploring Skin Softness Perception and Virtual Body Embodiment Techniques to Enhance Self-Haptic Interactions

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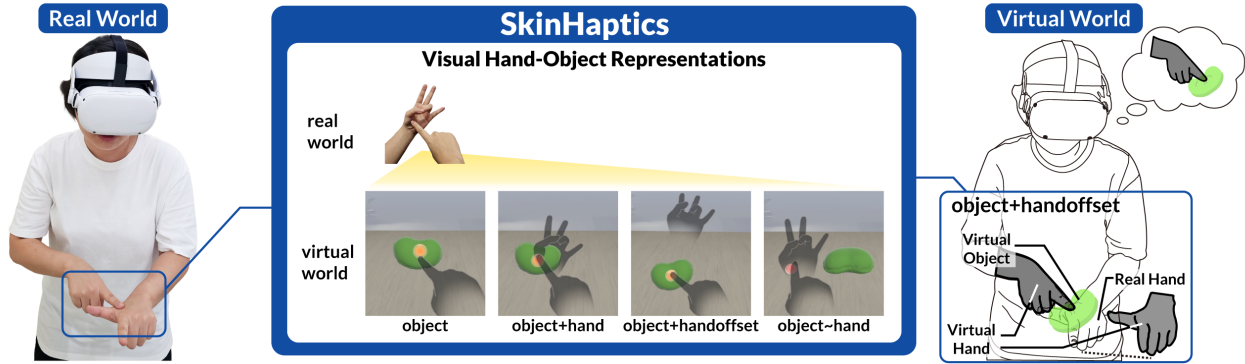


Figure 1: A user wearing a head-mounted display touches a virtual object in VR. SkinHaptics, a self-haptic interface methodology for providing haptic effects on soft deformable objects, allows the user to experience varying softness of the object using their own hand.

Abstract

Providing haptic feedback for soft, deformable objects is challenging, requiring complex mechanical hardware combined with modeling and rendering software. As an alternative, we advance the concept of self-haptics, where the user's own body delivers physical feedback, to convey dynamically varying softness in VR. Skin can exhibit different levels of contact softness by altering the biomechanical state of the body. We propose SkinHaptics, a device-free approach that changes the states of musculoskeletal structures and virtual hand-object representations. To demonstrate this methodology, we designed interaction scenarios with SkinHaptics.

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CCS Concepts

• **Human-centered computing** → **Virtual reality**; Haptic devices.

Keywords

Softness, Haptics, On-body interaction, Interaction Techniques, Embodiment, Virtual Reality

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1 Introduction

Significant effort has been made within the research community to provide realistic and sophisticated haptic feedback for immersive experiences in virtual environments. One area of focus has been the

development of encounter-type haptic displays, which offer a tangible surface for haptic interactions [6]. These displays allow users to actively engage with surfaces that have programmable shape, kinesthetic feedback, and tactile properties [2, 6]. Researchers have designed and validated advanced encountered-type displays that emphasize various aspects, such as surface reconstruction [8, 12, 13], surface actuation mechanisms [9, 14, 16], and mobility [1, 4, 8]. Despite their realistic and engaging haptic experiences, it is not easy to reach out these high-end devices in many practical use scenarios. This is largely due to their technical complexity and cost. Additionally, even with such sophisticated devices, rendering high-resolution softness feedback for soft, deformable virtual objects remains a challenge, particularly when human perception [15] and haptic rendering frameworks [10, 11] are not fully considered.

In this paper, we introduce SkinHaptics, a mobile and on-demand self-haptic interface methodology for providing haptic feedback on soft, deformable objects, extending the concept of self-haptics. The term “self-haptics” refers to engaging the parts of one’s own body as a means of delivering haptic feedback in virtual reality (VR) [3]. We aim to improve and solidify this intriguing concept by filling the research gaps that we identified in self-haptics. The primary concern with self-haptics may be that users are aware that they are touching their own bodies. To address this, we design and evaluate virtual body representations to control the embodiment of the self-haptic body. Compared to previous self-haptic approaches, SkinHaptics is designed to support rich haptic interaction scenarios that require 1) varying multi-level softness, 2) a relatively wide range of softness difference, and 3) less perceptible self-touch sensation.

2 Design of SkinHaptics

We formulate a design space for SkinHaptics, a self-haptic methodology for the non-dominant hand, based on our understanding of skin characteristics, human softness perception, and visuo-haptic interactions.

For self-haptic interaction, we need to determine the body part to use as a passive prop. The user must be able to move the body part easily to the interaction position in a specified orientation. It is also advantageous if the body part can present the softness varying widely. Obviously, the hand is the best organ that satisfies these two conditions, contributing to rich self-haptic effects. In SkinHaptics, as depicted in Figure 1, the user uses the dominant hand for interaction and the non-dominant hand as a passive prop providing haptic feedback, possibly with audiovisual stimuli that can enhance the embodiment of virtual objects. This use scenario offers a few design variables, and we delve into them in what follows.

2.1 Gesture and Contact Point

The hand is a versatile organ with a complex musculoskeletal structure (e.g., bones, muscles, and tendons) in a confined volume. The hand in a different gesture modulates its viscoelastic property, which also depends on the contact point. To perform gestures, some finger muscles must contract or relax, which causes the finger(s) to move. These muscle contractions cause the overlying skin to become stiff. Thus, the gesture and contact point, as a combination, are an important design variable to deliver haptic feedback of varying softness sensations.

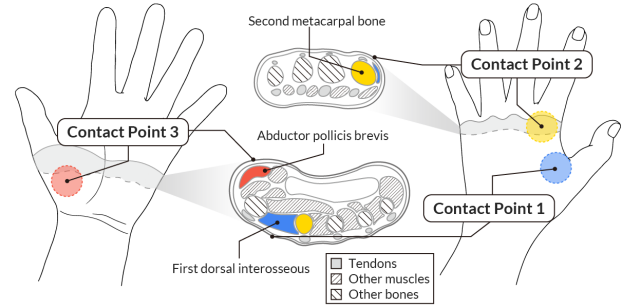


Figure 2: Three contact points on the hand. They are located on the first dorsal interosseus, second metacarpal bone, and abductor pollicis brevis, respectively. The musculoskeletal system of the hand, including these points, is illustrated in cross-sectional diagrams.

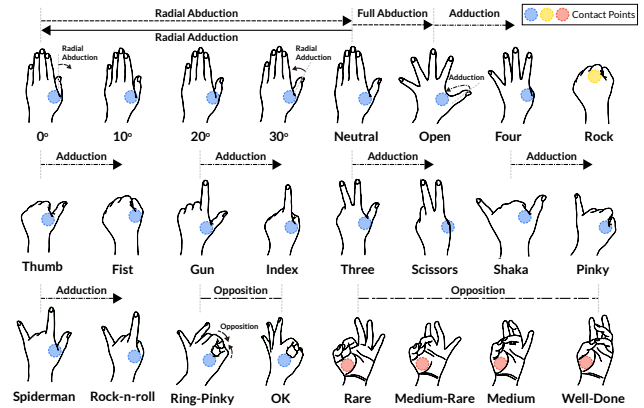


Figure 3: 24 hand gestures with contact points. Circles are colored to indicate different contact points (Blue: Contact Point 1, Yellow: Contact Point 2, Red: Contact Point 3). Thumb movements include radial abduction, radial adduction, full abduction, adduction, and opposition. The directions of the thumb movements are indicated by arrows at 0° (radial abduction), 30° (radial adduction), Open (adduction), and Ring-Pinky (opposition).

We choose to use three contact points based on their distinct musculoskeletal structures, as illustrated in Figure 2. Contact Points 1 and 3 are located at the muscle belly of the first dorsal interosseus (FDI) and the abductor pollicis brevis (APB), respectively. Contact Point 2 is on the second metacarpal bone, which involves a joint rather than muscle tissue under the skin unlike the other two. These three contact points are combined with one-handed gestures. We carefully select socially acceptable gestures that are free of any awkward or overly tense movements, as shown in Figure 3. They were motivated by thumb tilt, number expression, common gestures, and steak doneness tests. These 24 gestures, all associated with contact points, are the key design choices in SkinHaptics.

2.2 Haptic Interaction Mode

For haptic interactions with SkinHaptics, we basically consider the following two interaction modes: embodied object and embodied device experience. In the embodied object mode, the non-dominant hand becomes a virtual object, and the object is attached to the hand when its pose matches a pose assigned to the object. Whenever the user changes the pose of the self-haptic hand, the object corresponding to the new pose is aligned and attached to the contact point of the hand. Second, the embodied device mode refers to interaction scenarios where users conceive of their non-dominant hand as a haptic interface that receives haptic feedback from remote objects.

2.3 Haptic Feedback Type

The haptic feedback provided by SkinHaptics can be categorized into three types, serving as another design variable.

First, SkinHaptics can be used to represent various passive props with a hand. At each contact point, it can provide static passive feedback from a fixed pose. It is allowed to change poses (or contact points) for haptic interactions with another object. We propose an additional scenario of using hand-worn objects like a smartwatch (watch face and strap), rings, and gloves to cover the physical haptic properties not covered in the hand (Figure 5(a)). It is feasible to modulate the haptic feedback of different virtual passive objects by allowing multiple contact points for a fixed pose (Figure 5(a) and (d)).

Second, SkinHaptics operates as a method providing passive dynamic haptic feedback. To support dynamic haptic feedback, the users can change hand poses while maintaining contact (Figure 5(b) and (e)). We can design a sequence of one-hand gestures that would create the stiffness curve of a deformable object for presenting continuously-changing stiffness at one contact point for the virtual object.

Third, we introduce a novel interaction scenario with pose constraint mechanisms. Unlike conventional self-haptic methods, such pose constraints can be enforced to make SkinHaptics programmable. As illustrated in Figure 5(c) and (f), external robotic mechanisms such as exoskeletons or kinesthetic haptic gloves can be used to constrain hand poses, which in turn control the softness of a contact body (e.g., Contact Point 1).

2.4 Hand-Object Representation

For self-haptic interactions, we design hand-object representations that are intended to reduce or increase the sense of embodiment. For the baseline, we visually overlay a virtual object over the hand (as in Hand Interfaces [7] and Marichal et al. [5]; see Figure 4(c)). In this representation, virtual objects are attached to the self-haptic hand. The design rationale for this is to induce the embodiment of the hand as an object [7].

We introduce the virtual representations designed to reduce virtual body embodiment, which is supposed to minimize the sense of self-touch. We first decouple the object from the baseline, and only this visualization works with the embodied device mode (see Figure 4(e)). Contrary to the previous one, we introduce an offset between the position of the real hand and its virtual representation (Figure 4(d)). The position of the virtual object follows the position of the real hand. This representation introduces an intentional

visual mismatch of not touching the virtual hand while touching the virtual object coupled to the real hand. We also hide the virtual hand to further reduce the embodiment of the self-haptic hand (Figure 4(b)). This object only representation can also be counted as an infinite hand-object offset.

3 Demonstration of SkinHaptics

SkinHaptics offers significant potential for a wide range of haptic applications in VR. We highlight bare-hand interaction scenarios with SkinHaptics in terms of the three following haptic feedback types: passive static, passive dynamic, and programmable.

3.1 Demo #1: Passive Feedback

The passive static type (Figure 5(a)) provides tactile sensations for static objects. This demonstration seeks to allow the user to experience the softness of different objects. In this demonstrated scenario, users play with multiple toys on the desk. Users can interact with toys such as dolls, guns, and cars and perceive the softness of interacting objects at the same time. Likewise, in a VR scenario where users are shopping in a virtual mall, they can press and compare many shoe insoles to identify the ones that best match their preferences. This type works effectively in scenarios where multiple objects have different but homogeneous materials.

3.2 Demo #2: Passive Dynamic Feedback

The passive dynamic type (Figure 5(b)) is designed to present the varying softness of an object (or UI). The VR scenarios of this type are tightly coupled with gesture recognition. We demonstrate a usability scenario where the radial position of the thumb is adjusted to control volume. In this scenario, the user can also haptically feel the change in volume by touching the skin over the FDI. Additionally, we can imagine a VR game scenario. In a VR game scenario, when a user's Index gesture is detected, a virtual bow appears on the non-dominant hand. Assuming that the bow is controlled by the abduction of the index finger, the user can move the index finger away from the center to draw the bowstring. The user can then feel the tension of the bowstring by touching the skin over the FDI muscle with the other hand.

3.3 Demo #3: Programmable Feedback

Third, we introduce a novel interaction scenario with pose constraint mechanisms. Unlike conventional self-haptic methods, such pose constraints can be enforced to make SkinHaptics programmable. As illustrated in Figure 5(c) and (f), external robotic mechanisms such as exoskeletons or kinesthetic haptic gloves can be used to constrain hand poses. The external device shown in Figure 5(c) not only assists to maintain constrained poses but also enables the representation of pre-programmed material properties. For example, in voxel modeling tools like Minecraft, users can feel different voxel softness by touching the programmable self-haptic hand. This brings haptic capability to shape modeling tools.

4 Conclusion

In this paper, we present SkinHaptics, a novel and device-free haptic methodology to facilitate self-haptic interactions through empirical

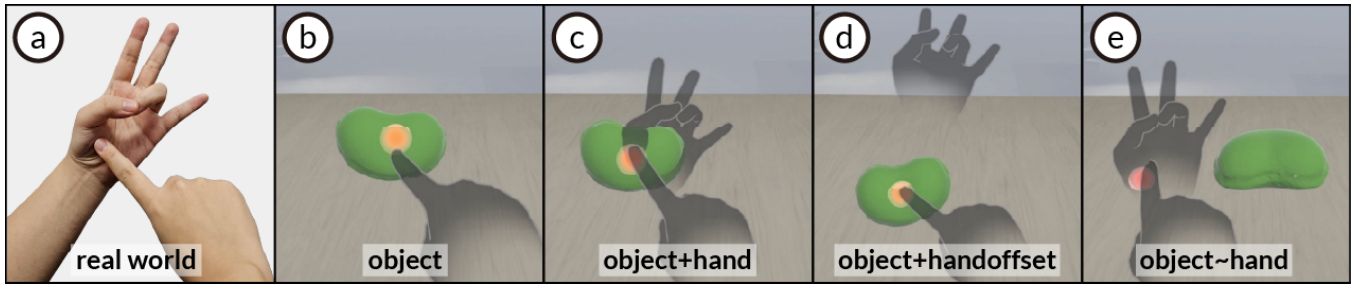


Figure 4: Example of a hand gesture and four corresponding visual hand-object representations. (a) the hand gesture in real world, (b) object, (c) object+hand, (d) object+handoffset, and (e) object~hand.

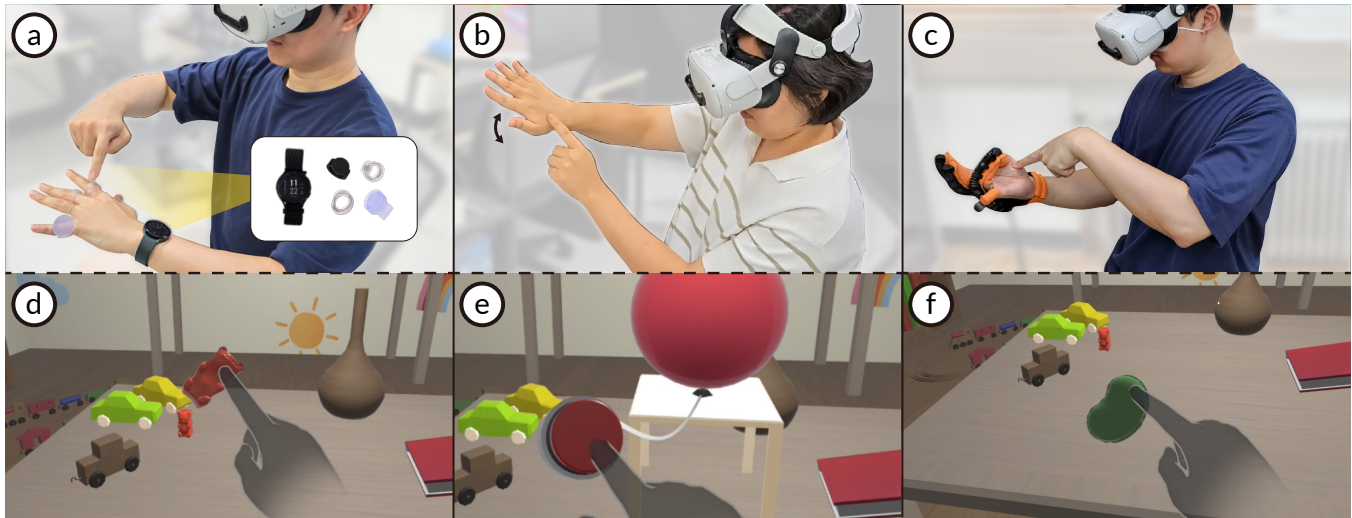


Figure 5: Use scenarios for SkinHaptics. Passive haptic feedback: (a) and (d). A user wearing 3D-printed rings and a smartwatch plays with multiple toys on the desk. In the real world, three 3D-printed rings (Shore hardness A 40, 70, and 80), Rock and Neutral SkinHaptics (Shore hardness OO 67 and 15), and a smartwatch face (rigid body) with a strap (Shore hardness OO 42) can be respectively mapped to virtual toys. Passive dynamic haptic feedback: (b) and (e). The user presses the air pump button to inflate the virtual ball and is guided to move the thumb toward or away from the palm to simulate the varying softness of the pump button depending on the air pressure of the ball. Programmable haptic feedback: (c) and (f). When the user selects the palm rest on the desk, a kinesthetic haptic glove automatically constrains the hand pose to a pre-defined gesture, allowing the hand to act as a self-haptic proxy.

evidence. Also, we demonstrate interaction scenarios with SkinHaptics. Our demonstrations suggest SkinHaptics significant potential for a wide range of haptic applications in VR.

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